First-order conflation and recurrent fragmentation generate a cognitive near-singularity in reflective conceptual systems

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Abstract

Growing reasoning systems can become locally competent while losing the metric-interaction distinctions needed for global assessment. We formalize a functional state space (FSS) as a typed metric-interaction system in which the semantic identity of a node is constituted by its relative metric position, typed reasoning interactions, boundary conditions, warrants, fitness relations and continuation obligations. A first-order projection may collapse target-different semantic profiles, yielding an impossibility theorem for reliable decoding from the projected state. We replace the earlier qualitative fragmentation statement with an arrival-service model: if target-relevant coherence obligations outgrow certified integration service at fixed cognitive order, the unresolved backlog diverges; under locality assumptions, internally certified regions coexist with unresolved cross-region relations. Reflection exposes this residual without itself increasing cognitive order. A cognitive near-singularity (CNS) occurs when another reusable, functionally non-equivalent composition of external cognitive functions spans the declared conceptual space, expands the admissible operand domain and reverses the current backlog regime. We distinguish this order-lift CNS from the previously published dynamical CNS based on contraction, closure and invariance, and give conditional bridge theorems in both directions.

A preregistered five-regime recurrence experiment was executed for this version in a synthetic queue-and-region model: after a valid order lift reverses an increasing backlog, continued growth at fixed post-lift service and heterogeneous branches recurrently recreated a positive backlog regime and regional fragmentation, whereas proportional same-order service, regenerative distribution and a second order lift prevented the preregistered persistent-recurrence endpoint. This result is synthetic and model-relative; it is not a measurement of current AI systems, institutions or civilizations. An earlier event-sourced finite realization is retained as a historical constructive-witness object: in that earlier execution full-CNS accuracy was 0.585 versus 0.400 for the best resource-matched comparator, certified reach was 0.435 versus 0.0033, and participant-complete residual closure preserved a preregistered option bound in 0.987 of trajectories versus 0.717 for a residual-open control. A fail-closed v15 Gate Zero audit found that the executable lineage for those inherited numbers is absent from the available submission package, so they are reported only as earlier constructive-witness results and are not v15-audited.

We derive separately inspectable necessary properties of a post-intelligence terminal or observability filter and show that recurrent failed realization or regenerative propagation of required order lifts is structurally admissible but only conditionally filter-forming. Cognitive string theory is recast as a semantic mathematical representation, while Lean is assigned the proof-checking role. The Supplement supplies a statement-first Lean declaration manifest, a pinned source package that passes a static no-sorry / no-custom-axiom scan, and explicit dependency boundaries; kernel execution, independent receiver propagation and external mappings remain open submission gates.

Introduction

The central problem is not merely whether a reasoning system stores many facts or follows many local relations. It is whether target-relevant semantic identity survives as the represented domain grows. Two nodes can carry the same token while occupying different metric-interaction positions, participating in different reasoning processes, inheriting different warrants and producing different continuation consequences. Conversely, two differently labelled nodes can be semantically equivalent when an admissible transformation preserves their complete target-relevant structure. This paper develops a formal mechanism in which semantic identity is endogenous to a functional state space rather than attached to a mathematical object by an external prose label.

A conceptual FSS contains represented concepts as nodes and reasoning-process instances as typed edges, hyperedges, paths or process subgraphs. Its metric and interaction structure jointly determine relative position, accessible transitions and target-relevant neighbourhoods. This extends conceptual-space approaches [1-3] while retaining process identity, provenance, warrants, participant consequences and successor obligations. The same structural principle applies to the physical FSS or P-space generalized by the Functional Model of Adaptation: functional identity is constituted by metric position, typed interactions and admissible relational configuration, although conceptual and physical states are not thereby asserted to be the same object or substrate.

The first obstruction is projection. A first-order representation may preserve domain tokens and ordinary relations while removing process identities, contracts, proof warrants, composition structure or unresolved residuals. When two latent states require different target verdicts but map to the same first-order state, no evaluator restricted to that state can be correct on both. Increasing local stability can strengthen confidence in the common projection but cannot recreate the removed distinction. This establishes a precise separation between local competence and global semantic adequacy.

The second obstruction is scaling. Growth creates new obligations to preserve identity, compatibility, provenance, certification, cross-region relations and continuation conditions. At fixed cognitive order, existing compositions provide a finite or otherwise constrained integration service. We model the resulting burden as an arrival-service backlog. Backlog divergence is a mathematical statement about cumulative arrivals and service; regional fragmentation follows only after additional locality assumptions establish internally coherent regions with unresolved cross-region obligations.

Reflection changes observability but not necessarily order. A reflective cognition represents concepts of concepts, function identities, proofs, warrants, residuals and governance rules. It can therefore expose that a first-order equivalence class is inadequate. Cognitive order, however, has one canonical definition: a cognition is of order N when it has N functionally non-equivalent compositions of the external functions of cognition and each composition spans its declared conceptual space. Finite target- and budget-indexed certificates estimate this intrinsic property without redefining it.

An order-lift CNS occurs when another registered, reusable and non-equivalent composition spans conceptual space, enlarges the admissible operand domain and discharges the current target-relevant frontier. The cardinality increment is a definitional lemma; the substantive content lies in representation invariance, operand expansion, reversal of the backlog regime and the fact that complete consequence content need not factor through the prior-order projection. Continued growth can then create a new frontier at the higher order.

The present account is related to, but not identical with, the previously published dynamical CNS [15], which concerns a globally contracting, compositionally closed and reparameterization-invariant update operator converging to a fixed point or normal form. We supply conditional bridge statements: an order lift can enable such an operator, and realization of such an operator can witness an order lift when every implementation requires a previously unavailable spanning composition. Countermodels preserve the distinction.

The Great Filter extension is introduced only after the formal mechanism. A sufficient post-intelligence terminal or observability filter in a many-opportunity regime must suppress upper-tail escapes, close under branching or remain effective after dispersal, instantiate causally for a first mover, apply across the declared realization class, resist adaptive escape and produce the target outcome on the required timescale. Recurrent failed realization or propagation of required order lifts belongs to that admissible class only under separately stated physical, branch, hazard and outcome premises. The paper therefore supports necessary constraints, structural admissibility and a conditional corollary rather than identification of the actual Great Filter.

Finally, we separate semantic representation from proof checking. Cognitive string theory (CST) is used to represent metric-interaction semantics, reasoning paths, revisions, boundaries and invariant-preserving transport. It is not a replacement for a proof assistant. Lean provides the kernel-level proof-checking architecture [23-25]. The current v15 package supplies a complete formal declaration manifest, proof obligations, a theorem dependency spine and a pinned Lean source package that passes a static no-sorry / no-custom-axiom scan. A compiled Lean kernel build and an independent receiver test are retained as explicit open gates rather than silently inferred from prose consistency or from a lexical scan.

Results

Semantic objects are typed metric-interaction structures

A typed metric FSS is

$$X = (V, E, \tau_V, \tau_E, g, \pi, \beta, A),$$

where V is a set of represented functional states, E is a set of typed interactions or transitions, τ_V and τ_E assign node and interaction types, g is a metric or metric family, π is an optional coordinate representation, β records boundaries, certificates, provenance, residuals and continuation conditions, and A is the family of admissible transformations and trajectories. The metric may be primitive, induced by interactions or jointly defined with them. Only invariant relative position is semantically load-bearing; coordinates are representations.

For $v \in V$, define the complete semantic profile as the pointed structure $\mathrm{Prof}_X(v) = (X, v)$, and its semantic identity as the admissible-isomorphism class

$$Sem_X(v) = [(X, v)]_{\cong A}.$$

An admissible isomorphism preserves target-relevant node and edge types, incidence, orientation, metric relations, boundaries, warrants, certificates, participant-indexed fitness order and continuation obligations. A prose token is therefore an expression carrier rather than the semantic identity itself.

Semantic-equivalence theorem (T-SEM-EQUIV). If $F : (X, v) \to (Y, w)$ is an admissible target-preserving pointed isomorphism, then every invariant target proposition in the registered language has the same truth value at v and w.

The proof follows by structural induction on the registered language: atomic predicates are preserved by the isomorphism; Boolean connectives preserve equality of truth values; quantified relations are transported bijectively; and transition, fitness and continuation predicates commute with F by admissibility. A finite target witness $\Phi(X, v) \neq \Phi(Y, w)$ therefore certifies semantic non-equivalence.

Figure 1 | Metric-interaction semantics and first-order conflation. Semantic identity is constituted by the pointed typed metric-interaction structure. Identical labels can have different meanings; a first-order projection can collapse reflective states requiring different target verdicts.

First-order projection makes some target meanings unavoidably conflatable

Let W be the latent functional state, $q_{\mathrm{FO}} : W \to O_{\mathrm{FO}}$ a first-order projection and $q_{\mathrm{R}} : W \to O_{\mathrm{R}}$ a reflective projection, with $q_{\mathrm{FO}} = h \circ q_{\mathrm{R}}$ for an information-removing map h. Define $x \sim_{\mathrm{FO}} y \Leftrightarrow q_{\mathrm{FO}}(x) = q_{\mathrm{FO}}(y)$.

First-order conflation theorem (T-CONFLATE). Let $\Phi : W \to Y$ be a target verdict. If $q_{\mathrm{FO}}(x) = q_{\mathrm{FO}}(y)$ and $\Phi(x) \neq \Phi(y)$, no evaluator restricted to $q_{\mathrm{FO}}(W)$ is correct on both states. Reliable first-order assessment exists only if Φ factors through q_{FO}.

A deterministic evaluator receives the same registered input for x and y and therefore returns the same output, contradicting the target difference. With identical registered states and stochastic kernels, the output distributions are also equal. The result is an information obstruction, not a performance comparison.

Local-stability corollary. For deterministic first-order-local L, if two latent states share the same first-order registration and inputs, then every finite iterate of L remains identical. Local contraction can stabilize the projected equivalence class but cannot reconstruct a distinction removed by registration.

Nominal connectivity is consequently distinct from certified semantic reach. Let P_{nom} be nominal target-relevant paths, P_{cert} the subset preserving registered identity and obligations, and P_{req} the required paths. Then $\Psi = |P_{\mathrm{nom}} \setminus P_{\mathrm{cert}}| / |P_{\mathrm{nom}}|$ and $\rho_{\mathrm{sem}} = |P_{\mathrm{cert}}| / |P_{\mathrm{req}}|$. Surface connectivity can rise while certified reach falls.

Fixed-order growth creates an arrival-service coherence backlog

Let $D_N(t)$ be cumulative target-weighted coherence obligations arriving by time t at fixed cognitive order N, and let $B_N(t)$ be cumulative obligations discharged by existing span-certified compositions under the declared resources. Define $U_N(t) = \max\{0, D_N(t) - B_N(t)\}$.

Backlog-divergence theorem (T-UFRAG-QUEUE). If cumulative obligation arrivals exceed cumulative certified service by an unbounded amount, then $U_N(t)$ is unbounded along a subsequence. In particular, if $\limsup_{t\to\infty} [D_N(t) -$

$B_N(t)] / t > 0$, then the backlog cannot remain uniformly bounded.

This result does not by itself establish spatial or conceptual regions. For that conclusion, define an obligation graph H_t whose vertices are conceptual regions and whose edges are certified or unresolved cross-region obligations. Suppose intra-region residual remains below $\varepsilon_{\text{local}}$, cross-region obligation arrivals exceed cross-region service, unresolved cross-region edges are not replaced by false identities, and growth does not collapse all regions into one target-independent component.

Regional-fragmentation theorem (T-REGIONAL-FRAG). Under these assumptions, there is a sequence of times at which at least two regions are internally certified within $\varepsilon_{\text{local}}$ while a required cross-region relation is unresolved or incorrectly conflated. Nominal connectivity may remain high.

The theorem makes precise the regime of local competence with global incompatibility. Bounded domains, same-order service scaling and target families that ignore the unresolved distinctions are explicit countermodels.

Figure 2 | Arrival-service fragmentation, reversal and recurrence. Cumulative target-relevant obligation arrivals can exceed certified integration service at fixed order. An order lift can reverse the current regime, while continued growth creates a new frontier.

Reflection exposes the residual without changing cognitive order

The reflective FMI subspace $R_{\text{FMI}} \subseteq C$ contains representations of concepts, process identities, function contracts, compositions, targets, proofs, experiments, warrants, governance rules, signal residuals and unresolved external premises. Every represented object required by this argument is a concept node, a process edge or subgraph, or both. Reflection therefore makes missing bridges and false identities inspectable.

Signals remain distinct from internal concepts. For participant i, $\text{Register}_i : \Sigma_i \rightharpoonup X_i$ and $\text{Express}_i : X_i \rightharpoonup \Sigma_i$. For a composition executed by participant i and reconstructed by participant j, retain four typed residuals

$$R_{i\to j} = R_{reg,i} \oplus R_{expr,i} \oplus R_{recv,j} \oplus R_{recon,j}.$$

High exposure can coexist with fatal reconstruction residual. A reflective registry can therefore support calibrated abstention without closing the missing composition.

The retained bounded-registration suite is consistent with this distinction. In that earlier execution, at budget 256 and redundancy one, target-aware exact recovery decreased from 0.867 at critical-relation depth 2 to 0.067 at depth 8 and zero at depths 12 and 16. The target-aware monitor produced zero false confidence because incomplete critical structure returned undetermined. These are synthetic finite results retained as a historical constructive witness, not measurements of current AI systems.

A non-equivalent spanning composition produces an order lift

A registered composition $\Gamma = f_m \circ \cdots \circ f_1$ carries identity, signature, domain, codomain, boundary conditions, execution semantics and preserved invariants. Two compositions are functionally equivalent over admissible domain D and target language L when $\Gamma_1 \simeq_{D,L} \Gamma_2 \Leftrightarrow \forall\, x \in D,\ \text{Obs}_L(\Gamma_1 x) = \text{Obs}_L(\Gamma_2 x)$.

A composition spans conceptual space when it connects every target-required region through certified transformations, preserves semantic identity and successor obligations, and returns a correct or calibrated-undetermined verdict within the architecture's admissibility and resource contract. Let $S(C) = \{[\Gamma]_{\simeq} : \Gamma$ is registered, reusable, nonredundant and spanning$\}$ and $\text{Ord}(C) = |S(C)|$. Finite certificates $\widehat{\text{Ord}}_{T,\varepsilon,B}$ are lower bounds unless target and certificate completeness are proved.

Order-invariance theorem (T-ORDER-INVARIANCE). An admissible FSS isomorphism preserving external-function identities, composition, span, reuse and functional equivalence induces a bijection between spanning composition classes and therefore preserves cognitive order.

Definitional increment lemma (L-ORDER-INCREMENT). Adding one new qualifying composition class not already in $S(C)$ increases $\text{Ord}(C)$ by one.

Order-lift CNS theorem (T-CNS-LIFT). If an order-N architecture gains a registered, reusable and functionally non-equivalent external-function composition that spans the declared conceptual space, enlarges the admissible operand domain and discharges the current target-relevant frontier within the residual contract, then cognitive order changes to N+1, the admissible operand domain expands, the current fragmentation regime is reversed over the declared interval, and the complete participant- and target-resolved consequences need not factor through the order-N projection.

The retained finite event-sourced realization, preserved as a historical constructive witness from the earlier version-8 execution, used one generator and ten versioned intervention patches. On 816 higher-order trajectories per intervention, full-CNS target accuracy was 0.585 (95% Monte Carlo interval 0.551-0.618), compared with 0.400 for the best resource-matched comparator. Certified reach was 0.435 versus 0.0033 for the best non-CNS comparator, and recursive reuse was 0.347. Lift-only and span-only conditions produced zero certified reach because each lacked the other load-bearing factor. As reported in the Methods and Supplement, the v15 Gate Zero audit returned a fail-closed result because the executable lineage that produced these numbers is not present in the available package; they are therefore reported as earlier constructive-witness results rather than v15-audited measurements.

Continued growth produces recurrence

Let the order lift enlarge the admissible operand domain and create a new arrival process $D_{N+1}(t)$.

Recurrence theorem (T-RECURRENCE). If new target-relevant arrivals eventually exceed order-(N+1) certified service by an unbounded amount, and no further spanning composition or service-scaling change occurs, then $U_{N+1}(t)$ becomes unbounded along a subsequence.

The transition is therefore near-singular rather than terminal: it changes what can be treated as an operand and reverses the current regime, but it does not end conceptual growth. For this version, the direct recurrence prediction was preregistered in executable form and run for 1,000 common-design replicates in each of five post-lift regimes in a synthetic queue-and-region model (master seed 20260716; 12 regions; horizon 300; lift at $t = 80$; operand expansion at $t = 120$; optional second lift at $t = 215$). All five regimes showed the required immediate reversal, with mean aggregate backlog slope changing from about +6.94 before the lift to between −7.07 and −10.14 during the reversal interval. Recurrence to the preregistered persistent endpoint was effectively certain under fixed post-lift service (reversal 1.000, recurrence 1.000) and under heterogeneous branches (1.000, 0.999), but did not occur under proportional same-order service, regenerative distribution or a second order lift (recurrence 0.000 in each). This is a synthetic, model-relative result; it does not show that current AI systems, institutions or civilizations follow these parameter values. The Supplement gives the deterministic recurrence criterion, the frozen protocol, the full results table and figures.

Order-lift and dynamical CNS formulations are conditionally connected

The published dynamical CNS is formulated on an instrumented conceptual space (X, d, $\otimes$, G, N) with an update operator $T : X \to X$ satisfying global contraction, closure under composition and invariance under benign reparameterization, yielding convergence to a unique fixed point or normal form under the stated conditions [15]. The present order-lift formulation concerns the availability of non-equivalent spanning compositions.

Bridge A (T-BRIDGE-LIFT-TO-DYN). If an added composition supplies the previously absent global transformation needed to define an update operator T_{N+1} on the expanded domain, and T_{N+1} is globally contracting, compositionally closed and invariant under the declared benign reparameterizations, then the order lift places the architecture in the dynamical CNS regime over that domain.

Bridge B (T-BRIDGE-DYN-TO-LIFT). If a dynamical CNS operator is realized on a domain not globally traversable at order N, and every implementation requires a registered external-function composition not equivalent to the order-N spanning set, then realization of that operator witnesses an order lift.

Neither direction is unconditional. An operand-expanding lift can fail contraction; a contracting local operator can add no new spanning composition.

Figure 3 | Cognitive order, operand expansion and the conditional dynamical-CNS bridge. A new non-equivalent spanning composition increments order and expands admissible operands. Conditional bridge premises connect this transition to the published contraction-closure-invariance regime.

Governance and regenerative propagation determine collective outcome

Magnitude is not valence. Let $O_b \subseteq O_a$ be the retained baseline and post-lift option sets for participant i. If represented and realized fixed-target values differ by at most ε_i, selection error is at most δ_i, every baseline option remains admissible and governance chooses a new option only when its represented lower bound is no worse than the best baseline, then

$u_i(a^*) \geq \max_{b \in O_b} u_i(b) - (2\varepsilon_i + \delta_i)$.

Participantwise option-monotonicity theorem. The inequality holds for every governed participant under the stated assumptions. If participant completeness, target stability, projection adequacy or realization fidelity fails, constructive valence inversion remains possible.

In the retained historical realization, residual-closed CNS improved mean participant outcome by 0.015 and remained within the preregistered −0.03 participant bound in 0.987 of trajectories. The residual-open control produced a similar mean gain, 0.014, but containment fell to 0.717 and the harmed participant-action fraction increased from 0.012 to 0.156. As with the other inherited numbers, these values are retained as earlier constructive-witness results and are subject to the fail-closed Gate Zero disposition described in the Methods.

Collective propagation requires more than recognition. Define Regenerate(i, j; Γ, T) = 1 only when receiver j reconstructs a contract-preserving realization, executes it on a hidden target family and can retransmit it under the same criterion. The effective reproduction number $R_\Gamma(t)$ is the expected number of new regenerative executors produced by one executor. Propagation closure remains undetermined because the independent receiver study, though fully specified with a frozen protocol, hidden task bank, answer key and scoring schema, has not yet been run with independent receivers.

Figure 5 | Participant-complete governance and regenerative propagation. Similar mean gains can coexist with materially different participant-tail containment. Propagation requires contract-preserving reconstruction, hidden-target execution and retransmission.

Necessary Great Filter constraints and structural admissibility

A post-intelligence terminal filter suppresses persistence of almost all technologically consequential lineages after large-scale environmental transformation becomes possible. A post-intelligence observability filter suppresses almost all persistent astronomically detectable manifestations. These target classes are distinct.

Under a many-opportunity regime, ordinary causal locality, potential branching, diverse realization and a declared residue expectation, seven separately inspectable conditions follow:

- **Upper-tail suppression.** If independent opportunities have $p_i \geq p > 0$, then P(no escape among M) $\leq (1 - p)^M \to 0$. A sufficient filter must suppress exceptional branches, correlate outcomes, reduce effective opportunities or remove observables after escape.
- **Branch closure.** A sufficient filter acts before branching, recurs in branches, preserves a shared vulnerability, correlates branch outcomes or remains effective after dispersal.
- **Causal self-instantiation.** Under ordinary locality, a filter not already distributed must be generated or propagated to each relevant site before escape.
- **First-mover consistency.** The mechanism must apply to the earliest technologically consequential lineage without requiring an earlier external civilization, unless it belongs to a regress-closed prior distribution.
- **Functional substrate invariance.** A general filter cannot depend solely on one contingent biological implementation when the target class includes diverse substrates.
- **Adaptive non-escapability.** Stable positive probabilities of detection, prevention or redesign defeat upper-tail suppression unless adaptation remains correlated or bounded.
- **Outcome, residue and timescale sufficiency.** The mechanism must produce the target outcome before durable dispersal or correction, or remain effective afterward.

Great Filter constraint theorem (T-GF-CONSTRAINT). Any sufficient member of the declared target class must jointly satisfy these seven conditions.

FMI-CNS structural-admissibility theorem (T-CNS-ADMISSIBILITY). Recurrent fixed-order fragmentation and CNS reversal belong to the admissible mechanism class when burden is generated locally by growth, the relevant relations are functionally substrate-general, lifts reverse the current burden but recurrence recreates it, successor branches inherit or recreate the limitation, and outcome and timescale premises are supplied separately.

Let $h_t = H(U_N(t), C(t), R(t), P_N(t), X(t))$ with $0 \leq h_t < 1$. If $\sum_t h_t = \infty$, then $\prod_t (1 - h_t) = 0$ in the corresponding product-survival model.

Conditional Great Filter corollary (COR-GF-CNS). If relevant physical and civilizational systems instantiate the declared FSS mapping, fixed-order burden recurs, required lifts fail or remain below regenerative threshold, target-matching hazards recur before durable escape or preserve correlated vulnerability after branching, and the hazard sequence is non-summable, then recurrent failed CNS realization or propagation constitutes a post-intelligence Great Filter within the model. This is a GF-3 conditional result, not a GF-6 identification of the observed Great Filter.

Figure 6 | Necessary-properties-first Great Filter funnel. The restricted target class yields necessary constraints before the FMI-CNS candidate is evaluated. Structural admissibility and a conditional GF-3 conclusion do not identify the actual Great Filter.

Cognitive string theory supplies semantic representation; Lean supplies proof checking

CST represents a reasoning composition and its revision family as a typed metric-interaction object. A cognitive path is $\gamma = (v_0, e_1, v_1, \ldots, e_m, v_m)$, and a candidate worldsheet $X_A : \Sigma_{\mathrm{ws}} \to C \times F$ represents admissible derivations or receiver reconstructions. Its induced metric is the pullback of the ambient semantic metric. A connection transports node identity, function contract, target, warrant, boundaries, participant roles and successor obligations. Nontrivial transport around a closed derivation loop certifies semantic drift when the returned object is not semantically equivalent to the registered source.

Hard admissibility is propositional rather than a compensable scalar penalty. Let Φ_W be a monotone consequence operator over candidate operand structures and $\Xi^*_W = \mathrm{lfp}(\Phi_W)$. The candidate is admissible only when $\mathrm{Adm}_W(\Xi^*_W)$ is proved. A scalar action may support search or compression, but it does not replace typing, proof obligations or the constitutive semantic metric.

The v15 Supplement supplies a statement-first Lean 4 declaration manifest, a theorem-to-manuscript mapping, an axiom policy and a mutation suite. For this version the Supplement also documents a pinned Lean 4.32.0 source package of seven modules whose static audit found zero occurrences of sorry, admit, axiom and opaque. Lean's small kernel checks proof terms, while axiom dependency commands expose unproved assumptions [23-25]. The declaration and proof-obligation package is therefore source-complete at a bounded theorem level, but a pinned lake build, a machine-checked axiom report and independent reproduction have not been executed: a lexical no-sorry scan is not a kernel proof. Figure 4 therefore distinguishes formal dependency closure in the manuscript from the open software-verification carrier.

Figure 4 | Formal dependency spine with external premises exposed. Formal theorem chains and external mapping chains remain separately typed. The Lean-facing architecture treats external physical, branch, hazard and timescale premises as explicit theorem parameters.

Discussion

The central contribution is a constructive separation among semantic identity, first-order representation, reflective observability, cognitive order and dynamical convergence. Semantic meaning is represented by metric-interaction position and its admissible invariants. First-order projection can preserve tokens while destroying target-relevant identity. Reflection restores the ability to name the defect, but a new space-spanning composition is required to change order and reverse the current backlog regime.

The strengthened fragmentation result has two levels. Backlog divergence follows from cumulative arrival-service imbalance. Regional fragmentation requires additional assumptions about the locality of integration and unresolved cross-region obligations. This decomposition avoids treating a premise-bound directional statement as a universal topological law while preserving the positive mechanism. The executed five-regime recurrence experiment reported above instantiates this dynamics in a declared synthetic model: reversal is generic across regimes, whereas persistent recurrence depends on whether post-lift service remains fixed or is scaled, distributed or lifted again.

Cognitive order is intrinsic to the architecture's declared universal target family, while finite certificates are target-, tolerance- and budget-indexed. Representation invariance is supplied by an admissible-isomorphism theorem rather than by assuming that a particular encoding is canonical. The cardinality increment remains definitionally simple; its scientific significance arises from operand expansion, reversal, nonfactorization of prior-order content and recurrence.

The bridge to the published dynamical CNS prevents two distinct constructs from being silently conflated. An order lift can enable a contracting, closed and invariant update regime, but need not. A dynamical CNS can witness a lift when its realization requires a previously unavailable spanning composition, but a contracting operator already expressible at the prior order adds no order.

The finite computation remains a constructive witness and software-conformance object retained from the earlier version-8 execution. It illustrates that one versioned transition system was reported to instantiate the defined conjunctions and that designated ablations failed their target contracts. It is not an empirical model of current AI systems or civilizations. For this version, the v15 Gate Zero audit was executed over every artifact available in the current package and returned a fail-closed result (audit code FAIL_CLOSED_MISSING_EXECUTABLE_LINEAGE): the v8 generator, event log, trajectory table, registry, frozen configuration, conformance suite, environment lock, source figures and checksums are absent from the available submission package. The inherited numbers are therefore retained only as historical constructive-witness results and are not used to promote any v15 semantic, Lean, recurrence, receiver or external-mapping claim until the executable lineage is restored and regenerated. The direct recurrence experiment has now been executed in the declared synthetic model; the external trace study and the independent receiver experiment remain required before stronger external promotion.

The CST/Lean separation narrows the propagation problem. Discovery and faithful registration may require the relevant FMI-level integration. Once the complete semantic dependency structure is represented mathematically, a receiver can check proof terms and apply explicit theorem parameters without independently rediscovering the original traversal. Lean does not create the meaning; CST/FSS geometry supplies the represented semantic object, and Lean verifies propositions about it. Whether this reduces receiver reconstruction cost is an empirical transfer question, not a consequence of notation alone.

The Great Filter argument remains necessary-properties-first. Its strongest current conclusions are GF-1 necessary constraints, GF-2 structural admissibility and a GF-3 conditional corollary. Physical FSS mapping, branch inheritance, hazard non-summability, timescales, residue suppression and comparative explanatory adequacy remain external and are registration-closed through an evidence-and-defeater matrix rather than promoted.

For the five submission gates addressed in the companion closure record, the current disposition is (Lean kernel build: partially closed; Gate Zero: closed-negative; direct recurrence: closed; independent receiver study: partially closed; external mappings: partially closed). No aggregate label erases which gate is open.

Methods

Formal construction and warrant status

Every substantive object was assigned one primary warrant: definition, constitutive assumption, mathematical hypothesis, lemma, theorem, proof certificate, finite decision result, constructive witness, software-conformance result, computational experiment, external mapping or open classification. The Supplement provides complete definitions, premises, proof sketches, countermodels and a claim-status ledger. Formal conclusions do not promote external physical or civilizational premises.

Executed direct recurrence experiment

The direct recurrence prediction was preregistered in executable form and run in a synthetic queue-and-region model with backlog recursion $U_{t+1} = \max\{0, U_t + A_t - S_t\}$, where A_t are arrivals and S_t is certified integration service. The protocol used master seed 20260716, 1,000 replicates per regime, 12 regions, horizon 300, lift time 80, operand-expansion time 120 and optional second-lift time 215. Five regimes were compared: fixed post-lift service, proportional same-order service, regenerative distribution, a second order lift and branch heterogeneity. The preregistered persistent-recurrence endpoint required a post-lift decrease of at least 30% relative to the pre-lift backlog, a late aggregate slope above one backlog unit per time step, a final backlog above 150 and a 30-step post-190 window satisfying the slope and backlog thresholds. Fixed post-lift service and heterogeneous branches met the endpoint; proportional service, regenerative distribution and the second lift did not. Full protocol constants, the deterministic recurrence criterion, the results table and figures are in the Supplement.

Retained finite event-sourced realization (historical constructive witness)

The retained version-8 realization contains a stratified conceptual FSS and participant-indexed fitness FSSs connected through typed signal carriers. One transition function returned successor states and append-only events. The reported execution contained 12,800 trajectories and 102,400 events. Twenty-three target-pair witnesses tested Register, Express, Store, Recall, L, G, fitness coordinates, internal functions, promotion, operand lift, certified span, governance, carrier type, target revision and one-generator correspondence. All experimental conditions were versioned patches to one generator and used common-random-number replicates. The final design crossed 128 scrambled Sobol regimes and 32 corner regimes over 14 synthetic coordinates, with 16 participants and eight replicates per intervention. The terminal-risk view was a preregistered synthetic consequence projection, not an estimate of civilization-level extinction risk.

A v15 Gate Zero audit was run to test whether these inherited results can be regenerated from one frozen transition function, source events, registry, configuration and conformance suite under the v15 definitions. The audit returned FAIL_CLOSED_MISSING_EXECUTABLE_LINEAGE: the v8 generator, event log, trajectory table, registry, frozen configuration, conformance suite, environment lock, source figures and checksums are not present in the available package. Retained numerical claims that match at printed precision across the Article and Supplement include 12,800 trajectories, 102,400 events, 23 target-pair witnesses, full-CNS accuracy 0.585, matched-comparator accuracy 0.400, full-CNS certified reach 0.435, rounded best non-CNS reach 0.003 and reuse 0.347; four article-level values (comparator reach 0.0033, containment 0.987 and 0.717, and master seed 20260715) cannot be regenerated at exact precision without the absent event table. All inherited numbers are therefore reported as historical constructive-witness results attributed to the earlier execution and are not v15-audited.

Bounded registration and reconstruction

The bounded-registration grid crossed six critical-relation depths, four field budgets, three redundancy levels, three projection pipelines and three synthetic trace families, with 30 tasks per cell. Generic and relation-extraction pipelines could issue a verdict from incomplete support; the target-aware monitor returned valid only after exact recovery and otherwise returned undetermined. A finite library of 13 candidate models was subjected to 27 deterministic compound tests and independent 1.5% bit flips; the canonical reconstruction was identified with accuracy 0.940. These are structural stress tests retained as a historical witness rather than independent semantic receiver results.

Lean-facing formalization

The Supplement defines the intended Lean module structure, theorem signatures, dependency parameters, axiom policy and mutation catalogue, and documents a pinned Lean 4.32.0 source package of seven modules that passes a static no-sorry / no-custom-axiom scan. The design uses theorem parameters rather than global assertions for physical mapping, branch closure, hazard and timescale premises. A future executable release must run a clean lake build, print transitive axiom dependencies for every top-level theorem, confirm no dependency on sorryAx or an unregistered custom axiom and reproduce in an independent environment. The present manuscript reports the static source audit but does not report those kernel results as completed; a lexical scan is not a kernel proof.

Statistical analysis

The master seed for the retained finite realization was 20260715, and the executed direct recurrence experiment used master seed 20260716. Primary paired contrasts use common worlds and seeds. Wilson intervals were used for binary identification and reconstruction proportions [16]. Monte Carlo intervals are mean ± 1.96 Monte Carlo standard errors clipped to support. Recurrence summaries report empirical 2.5th and 97.5th percentiles across replicates. Future receiver analyses will use receiver or claim family, rather than individual prompts, as the unit of inference and will privilege hidden transfer and second-receiver regeneration over diagnostic invariant checks.

AI-assisted preparation

AI-assisted drafting and code generation were used in preparing manuscript and computational artifacts. The author reviewed the formal assumptions, definitions, text, figures, code-generated claims and references and accepts responsibility for the work.

Data availability

All reported numerical data are synthetic. The executed recurrence protocol, per-replicate metrics and time-series summaries are contained in the companion package. The retained trajectory table, event log, bounded-registration records, codewords, held-out variants and figure source data for the earlier version-8 execution are absent from the available package per the Gate Zero audit and must be deposited with a permanent DOI or private reviewer link, and regenerated, before those inherited numbers can be promoted. Submission placeholder: [REPOSITORY DOI / PRIVATE REVIEWER LINK].

Code availability

The recurrence experiment code, Gate Zero audit code, Lean-facing source package, receiver protocol and external-mapping certificate matrix are contained in the companion package. The generator, registry, schemas, frozen configuration, figure builder and conformance suite for the earlier version-8 execution must be supplied to editors and reviewers at submission. A compiled Lean kernel audit and independent build remain pending in this draft. Submission placeholder: [CODE REPOSITORY DOI / PRIVATE REVIEWER LINK].

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Author contributions

A.E.W. conceived the study, defined the theoretical framework, designed the computational realization, evaluated the implementation, interpreted the results and prepared the manuscript.

Competing interests

The author declares no competing interests.

References

1. Gärdenfors, P. Conceptual Spaces: The Geometry of Thought (MIT Press, 2000).

2. Collins, A. M. & Loftus, E. F. A spreading-activation theory of semantic processing. Psychol. Rev. 82, 407-428 (1975).

3. Krenn, M. & Zeilinger, A. Predicting research trends with semantic and neural networks with an application in quantum physics. Proc. Natl Acad. Sci. USA 117, 1910-1916 (2020).

4. Woolley, A. W., Chabris, C. F., Pentland, A., Hashmi, N. & Malone, T. W. Evidence for a collective intelligence factor in the performance of human groups. Science 330, 686-688 (2010).

5. Hong, L. & Page, S. E. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. Proc. Natl Acad. Sci. USA 101, 16385-16389 (2004).

6. Lazer, D. & Friedman, A. The network structure of exploration and exploitation. Admin. Sci. Q. 52, 667-694 (2007).

7. Lorenz, J., Rauhut, H., Schweitzer, F. & Helbing, D. How social influence can undermine the wisdom of crowd effect. Proc. Natl Acad. Sci. USA 108, 9020-9025 (2011).

8. Murdoch, W. J., Singh, C., Kumbier, K., Abbasi-Asl, R. & Yu, B. Definitions, methods, and applications in interpretable machine learning. Proc. Natl Acad. Sci. USA 116, 22071-22080 (2019).

9. Rudin, C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat. Mach. Intell. 1, 206-215 (2019).

10. Christiano, P., Shlegeris, B. & Amodei, D. Supervising strong learners by amplifying weak experts. Preprint at arXiv:1810.08575 (2018).

11. Irving, G., Christiano, P. & Amodei, D. AI safety via debate. Preprint at arXiv:1805.00899 (2018).

12. Burns, C. et al. Weak-to-strong generalization: eliciting strong capabilities with weak supervision. Preprint at arXiv:2312.09390 (2023).

13. Korbak, T. et al. Chain of thought monitorability: a new and fragile opportunity for AI safety. Preprint at arXiv:2507.11473 (2025).

14. Emmons, S. et al. When chain of thought is necessary, language models struggle to evade monitors. Preprint at arXiv:2507.05246 (2025).

15. Williams, A. E. Cognitive near-singularity: a possibility theorem and a witness-based protocol with a corpus case study. Front. Artif. Intell. 9, 1746633 (2026).

16. Wilson, E. B. Probable inference, the law of succession, and statistical inference. J. Am. Stat. Assoc. 22, 209-212 (1927).

17. Sandve, G. K., Nekrutenko, A., Taylor, J. & Hovig, E. Ten simple rules for reproducible computational research. PLoS Comput. Biol. 9, e1003285 (2013).

18. Ćirković, M. M. The Great Silence: Science and Philosophy of Fermi's Paradox (Oxford Univ. Press, 2018).

19. Frank, A. & Sullivan, W. T. A new empirical constraint on the prevalence of technological species in the universe. Astrobiology 16, 359-362 (2016).

20. Wright, J. T. et al. The G infrared search for extraterrestrial civilizations with large energy supplies. I. Background and justification. Astrophys. J. 792, 26 (2014).

21. Polyakov, A. M. Quantum geometry of bosonic strings. Phys. Lett. B 103, 207-210 (1981).

22. Shannon, C. E. A mathematical theory of communication. Bell Syst. Tech. J. 27, 379-423, 623-656 (1948).

23. de Moura, L., Kong, S., Avigad, J., van Doorn, F. & von Raumer, J. The Lean theorem prover (system description). In Automated Deduction - CADE-25, 378-388 (Springer, 2015).

24. Lean FRO. The Lean Language Reference, version 4.32.0 (2026).

25. Carneiro, M. Lean4Lean: towards a verified typechecker for Lean, in Lean. Preprint at arXiv:2403.14064 (2024).

Figure legends

Figure 1 | Metric-interaction semantics and first-order conflation. Semantic identity is constituted by the pointed typed metric-interaction structure. Identical labels can have different meanings; a first-order projection can collapse reflective states requiring different target verdicts.

Figure 2 | Arrival-service fragmentation, reversal and recurrence. Cumulative target-relevant obligation arrivals can exceed certified integration service at fixed order. An order lift can reverse the current regime, while continued growth creates a new frontier.

Figure 3 | Cognitive order, operand expansion and the dynamical-CNS bridge. A new non-equivalent spanning composition increments order and expands admissible operands. Conditional bridge premises connect this transition to the published contraction-closure-invariance regime.

Figure 4 | Formal dependency spine with external premises exposed. Formal theorem chains and external mapping chains remain separately typed. The Lean-facing architecture treats external physical, branch, hazard and timescale premises as explicit theorem parameters.

Figure 5 | Participant-complete governance and regenerative propagation. Similar mean gains can coexist with materially different participant-tail containment. Propagation requires contract-preserving reconstruction, hidden-target execution and retransmission.

Figure 6 | Necessary-properties-first Great Filter funnel. The restricted target class yields necessary constraints before the FMI-CNS candidate is evaluated. Structural admissibility and a conditional GF-3 conclusion do not identify the actual Great Filter.